

Beyond the Unified Report:

Orbit-Uniform GCD Control, Cyclotomic Isolation, and a Covolume Budget Law for the Alaoglu–Erdős Conjecture

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Abstract

Assume hypothetically that a nonintegral real number x satisfies $M = 2^x \in \mathbb{Z}$ and $A = 3^x \in \mathbb{Z}$. The supplied unified report develops an extensive collection of equivalent formulations, determinant constructions, finite-place obstructions, rational secants, Kummer-theoretic diagnostics, and formalization modules. The present report deliberately does not repeat that material. It isolates and proves four further facts.

First, the Corvaja–Zannier S -unit theorem can be applied to one fixed rank-two group

$$\Gamma_x = \langle (2, M), (3, A) \rangle \subset \mathbb{G}_m^2(\mathbb{Q}),$$

which makes its generalized-gcd estimate uniform over *varying* 2–3 directions. This removes a fixed-direction restriction recorded in the supplied report. Second, at a prime interpolation order ℓ , almost all of the homogeneous cyclotomic factor $\Phi_{2\ell}(P, Q)$ survives not merely in an unreduced term but in the reduced denominator of the *entire* rational stencil. The same argument works for every maximally flat integer-node stencil whose largest node is ℓ . Third, an exact generating identity and a nearest-pole calculation give a variable-order asymptotic for these stencils. Combining it with the cyclotomic denominator theorem and a two-logarithm lower bound shows that a large class of linear extrapolation schemes produces approximants of quality only $H^{-o(1)}$ relative to their reduced denominators. Thus increasing the contact order does not repair the height mismatch. Fourth, a two-dimensional lattice calculation proves a sharp “covolume conservation law”: forcing a prescribed amount of p -adic divisibility into approximation denominators costs exactly the same factor in the guaranteed real approximation scale.

These results are rigorous progress, but they do not prove the conjecture. They close a broad linear-stencil route, sharpen the moving-direction gcd input, and leave three more focused interfaces: a nonlinear construction that escapes prime-top isolation, a genuinely output-specific local-to-global cancellation theorem, or a tensor-period theorem strong enough to control the quadratic logarithmic determinant.

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1 Proof status and nonduplication boundary

The conjecture is

$$2^x, 3^x \in \mathbb{Z} \implies x \in \mathbb{Z}. \quad (1)$$

The supplied report already contains, among much else, the rational-exponent reduction, the logarithmic determinant formulation, the four-exponentials barrier, exact conditional semigroup structure, rational-function rigidity, valuation lattices, determinant and interpolation families, rational secants, fixed-order Richardson extrapolation, Kummer support analysis, and a large formalization inventory. We use those conclusions as background and do not restate their proofs.

The additions proved here are the following.

New rigorous advances.

1. A fixed-group formulation makes the Corvaja–Zannier generalized-gcd bound uniform over all moving 2–3 directions generated by a hypothetical candidate.
2. A prime-order cyclotomic factor survives in the *reduced denominator after summing* a complete rational stencil. A local tax lemma also quantifies what arbitrary coefficients must pay to defeat this survival.
3. Every maximally flat integer-node linear stencil admits an exact generating function. Its variable-order error is controlled by the two nearest poles of $\sinh(xz)/\sinh z$.
4. Along convergent directions, all prime-top maximal-contact stencils in a natural growing window have approximation exponent $o(1)$ relative to their reduced denominator. This closes a broad linear-extrapolation architecture.
5. Prescribed p -adic denominator depth increases the approximation lattice covolume by the reciprocal factor, so the basic geometry-of-numbers architecture gives no adelic exponent gain.

No claim of a complete proof. No contradiction with the existence of a hypothetical nonintegral x is obtained. Every theorem below is either unconditional or is a theorem under the explicit temporary hypothesis that such an x exists. The remaining logical gap is identified in [Section 12](#).

1.1 Public-source audit of the competitive claims

Anthropic publicly launched Claude Fable 5 on June 9, 2026 [1]. Terence Tao subsequently gave a mathematical exposition of an explicit three-dimensional Jacobian-conjecture counterexample found using Fable [10]; Epoch AI provisionally recorded the reported Hadamard construction at order 668 and the completion of the previously unresolved orders below 2000 [5]. The ICARM Elliptic Curve Rank Leaderboard records a curve with thirty independent rational points, proving the unconditional lower bound $\text{rank } E(\mathbb{Q}) \geq 30$, and attributes the construction to Claude working with Levent Alpöge and Ava Howell [8]. This is a rank lower bound; exact rank 30 is not presently unconditional.

Targeted searches for a public Fable proof, preprint, repository, or Lean formalization of (1) did not locate one as of August 22, 2026. This does not contradict the user’s report of private work; it only means that no alleged new argument was available for mathematical inspection. The analysis below is therefore independent rather than a reaction to an unpublished proof sketch.

2 The hypothetical rank-two orbit

Assume for the remainder of the mathematical development that

$$x \notin \mathbb{Z}, \quad M = 2^x \in \mathbb{Z}, \quad A = 3^x \in \mathbb{Z}. \quad (2)$$

The elementary rational-exponent argument implies that x is irrational. Also $x > 1$: negative x gives $0 < 2^x < 1$, while $0 < x < 1$ gives $1 < 2^x < 2$, and neither interval contains a positive integer.

For $w = (a, b) \in \mathbb{Z}^2$ define

$$r_w = 2^a 3^b, \quad s_w = M^a A^b = r_w^x. \quad (3)$$

If $w \neq 0$, replace w by $-w$ when necessary so that $r_w > 1$; then $s_w > 1$. Write

$$r_w = \frac{P}{Q}, \quad s_w = \frac{U}{V} \quad (4)$$

in lowest positive terms. In particular, $P > Q$, $U > V$, and $h(r_w) = \log P$ for the absolute logarithmic height on $\mathbb{Q}$.

The central object is the fixed finitely generated group

$$\Gamma_x = \{(2^a 3^b, M^a A^b) : (a, b) \in \mathbb{Z}^2\} = \langle (2, M), (3, A) \rangle \subset \mathbb{G}_m^2(\mathbb{Q}). \quad (5)$$

The key point is that the direction w may vary while every point remains in this one group.

Lemma 2.1 (Multiplicative independence on every nonzero direction). *Under (2), r_w and s_w are multiplicatively independent for every $w \neq 0$.*

Proof. Suppose $r_w^c s_w^d = 1$ for integers c, d . Since $s_w = r_w^x$ and $r_w \neq 1$, taking real logarithms gives $c + dx = 0$. If $d = 0$, then $c = 0$. If $d \neq 0$, then $x = -c/d \in \mathbb{Q}$, which together with $2^x \in \mathbb{Z}$ forces $x \in \mathbb{Z}$, contrary to (2). $\square$

Lemma 2.2 (Equivalence of directional heights). *There is a constant $C_x \geq 1$, depending only on the hypothetical pair (M, A) , such that for all $w \in \mathbb{Z}^2$,*

$$C_x^{-1} h(r_w) \leq h(s_w) \leq C_x h(r_w). \quad (6)$$

Proof. The maps

$$(a, b) \mapsto h(2^a 3^b), \quad (a, b) \mapsto h(M^a A^b)$$

extend to piecewise-linear norms on $\mathbb{R}^2$. The first is nondegenerate because 2 and 3 are multiplicatively independent. The second is nondegenerate because a relation $M^a A^b = 1$ would, after taking logarithms and dividing by $x > 0$, give $2^a 3^b = 1$. All norms on the finite-dimensional vector space $\mathbb{R}^2$ are equivalent. $\square$

3 Orbit-uniform generalized-gcd control

Corvaja and Zannier prove that, for a fixed finitely generated group $\Gamma \subset \mathbb{G}_m^2(\overline{\mathbb{Q}})$, the generalized gcd of $u - 1$ and $v - 1$ is sublinear in the height, uniformly over multiplicatively independent $(u, v) \in \Gamma$ outside a finite exceptional set [4]. Applying their result to *the fixed group (5)*, rather than fixing one ray first, yields the following moving-direction statement.

Theorem 3.1 (Orbit-uniform homogeneous gcd bound). *Assume (2). For every $\varepsilon > 0$ there is $H_0 = H_0(x, \varepsilon)$ such that, for every nonzero direction w , every $n \geq 1$, and the reduced data (4), if*

$$n \max\{h(r_w), h(s_w)\} \geq H_0,$$

then

$$\log \gcd(P^n - Q^n, U^n - V^n) \leq \varepsilon n \max\{h(r_w), h(s_w)\}. \quad (7)$$

Consequently, uniformly along every sequence for which $n \log P \rightarrow \infty$,

$$\boxed{\log \gcd(P^n - Q^n, U^n - V^n) = o_x(n \log P)}. \quad (8)$$

The direction w is allowed to vary with n .

Proof. Set $u = r_w^n$ and $v = s_w^n$. By Theorem 2.1, u and v are multiplicatively independent, and $(u, v) \in \Gamma_x$. Corollary 1 of Corvaja–Zannier states that, for fixed Γ_x and every $\varepsilon > 0$, all but finitely many such independent pairs satisfy

$$\sum_{\mu} \log^- \max\{|u - 1|_{\mu}, |v - 1|_{\mu}\} > -\varepsilon \max\{h(u), h(v)\}. \quad (9)$$

At a finite prime p , a negative contribution occurs precisely when p divides both homogeneous numerators $P^n - Q^n$ and $U^n - V^n$; its magnitude is the minimum of their p -adic valuations times $\log p$. The archimedean contribution is nonpositive. Hence the left side of (9) is at most $-\log \gcd(P^n - Q^n, U^n - V^n)$. Also $h(u) = nh(r_w)$ and $h(v) = nh(s_w)$. This proves (7) once the height exceeds the maximum height of the finite exceptional set. Finally use Theorem 2.2 and $h(r_w) = \log P$. $\square$

Remark 3.2 (Why this is stronger than a fixed-ray use). For each fixed direction, the classical result gives an $o(n)$ estimate. Theorem 3.1 instead fixes the ambient group and lets the point move through all powers of all directions. The exceptional set is finite in the fixed group, so it cannot follow an unbounded sequence of changing directions. This is exactly the uniformity needed in the cyclotomic argument below.

4 Prime-order cyclotomic isolation

For an odd prime ℓ define the homogeneous cyclotomic factor

$$\Phi_{2\ell}(P, Q) = \Phi_{2\ell}(P/Q)Q^{\ell-1} = \frac{P^{\ell} + Q^{\ell}}{P + Q}. \quad (10)$$

Because $P > Q > 0$,

$$\log \Phi_{2\ell}(P, Q) = (\ell - 1) \log P + O(1), \quad (11)$$

with an absolute $O(1)$ term.

For a positive integer N , write $N_{(\ell)} = N/\ell^{v_{\ell}(N)}$ for its prime-to- ℓ part. Define the fresh surviving factor

$$\mathfrak{F}_{\ell}(P, Q; U, V) = \left(\frac{\Phi_{2\ell}(P, Q)}{\gcd(\Phi_{2\ell}(P, Q), U^{2\ell} - V^{2\ell})} \right)_{(\ell)}. \quad (12)$$

Lemma 4.1 (Order of every fresh prime). *Every prime $p \mid \mathfrak{F}_{\ell}(P, Q; U, V)$ satisfies*

$$p > 2\ell, \quad \text{ord}_p(PQ^{-1}) = 2\ell. \quad (13)$$

Moreover $v_{\ell}(\Phi_{2\ell}(P, Q)) \leq 1$.

Proof. A prime dividing (10) cannot divide PQ . For $p \neq \ell$, the standard cyclotomic order criterion gives $\text{ord}_p(PQ^{-1}) = 2\ell$. Therefore $2\ell \mid p - 1$, proving $p > 2\ell$. The prime 2 does not divide $\Phi_{2\ell}(P, Q)$ when P, Q are coprime and ℓ is odd. If $\ell \mid P + Q$, the lifting-the-exponent lemma gives

$$v_\ell(P^\ell + Q^\ell) = v_\ell(P + Q) + 1,$$

so the quotient in (10) has ℓ -adic valuation one. If $\ell \nmid P + Q$, the quotient is 1 modulo ℓ by Fermat's congruence. Thus $v_\ell \leq 1$. $\square$

Proposition 4.2 (Asymptotic mass of the fresh factor). *Under (2), uniformly over varying directions,*

$$\log \mathfrak{F}_\ell(P, Q; U, V) = (\ell - 1) \log P - o_x(\ell \log P) \quad (14)$$

as $\ell \log P \rightarrow \infty$ through odd primes.

Proof. The gcd in (12) is bounded by

$$\gcd(P^{2\ell} - Q^{2\ell}, U^{2\ell} - V^{2\ell}).$$

By Theorem 3.1, its logarithm is $o_x(\ell \log P)$ uniformly in the direction. Removing the ℓ -part costs at most $\log \ell$, by Theorem 4.1; this is $o(\ell \log P)$. Combine with (11). $\square$

4.1 Rational symmetric secants

For $k \geq 1$ set

$$J_k(r, s) = \frac{s^k - s^{-k}}{r^k - r^{-k}} = \frac{(U^{2k} - V^{2k})(PQ)^k}{(P^{2k} - Q^{2k})(UV)^k} \in \mathbb{Q}. \quad (15)$$

This is the rational value of $\sinh(xkh)/\sinh(kh)$ when $r = e^h$ and $s = r^x$.

Lemma 4.3 (Unique top-level negative valuation). *Let $p^d \parallel \mathfrak{F}_\ell(P, Q; U, V)$. If ℓ is larger than every prime divisor of MA , then*

$$v_p(J_\ell) = -d, \quad v_p(J_k) \geq 0 \quad (1 \leq k < \ell). \quad (16)$$

Proof. By Theorem 4.1, $p > 2\ell$ and $\text{ord}_p(PQ^{-1}) = 2\ell$. Thus $p \nmid PQ$, and p divides $P^{2\ell} - Q^{2\ell}$ only through the factor $\Phi_{2\ell}(P, Q)$. The definition of $\mathfrak{F}_\ell$ removes exactly the valuation canceled by $U^{2\ell} - V^{2\ell}$, which gives $v_p(J_\ell) = -d$. Since $p > 2\ell$ and ℓ exceeds the fixed prime support of MA , we also have $p \nmid UV$. For $k < \ell$, the order 2ℓ does not divide $2k$, so $p \nmid P^{2k} - Q^{2k}$. Therefore J_k is p -integral. $\square$

Theorem 4.4 (Reduced-denominator survival for a whole stencil). *Assume that ℓ is larger than every prime divisor of MA . Let $a_1, \dots, a_\ell \in \mathbb{Q}$ with $a_\ell \neq 0$, and suppose that for every prime $p \mid \mathfrak{F}_\ell(P, Q; U, V)$ one has*

$$v_p(a_\ell) = 0, \quad v_p(a_k) \geq 0 \quad (1 \leq k < \ell),$$

where $v_p(0) = +\infty$. Then

$$\mathfrak{F}_\ell(P, Q; U, V) \mid \text{den} \left(\sum_{k=1}^{\ell} a_k J_k(r, s) \right). \quad (17)$$

In particular, as $\ell \log P \rightarrow \infty$ through odd primes,

$$\log \text{den} \left(\sum_{k=1}^{\ell} a_k J_k(r, s) \right) \geq (\ell - 1) \log P - o_x(\ell \log P). \quad (18)$$

Proof. Fix $p^d \parallel \mathfrak{F}_\ell$. By [Theorem 4.3](#) and the coefficient hypotheses, the top summand $a_\ell J_\ell$ has valuation $-d$, while every lower summand is p -integral. The ultrametric inequality with a unique smallest valuation gives valuation $-d$ for the whole sum. Multiply over all primes dividing $\mathfrak{F}_\ell$, then apply [Theorem 4.2](#). $\square$

This is stronger than an unreduced-denominator estimate: it rules out cancellation *after all levels of the stencil have been added*.

4.2 A coefficient tax without a support hypothesis

Arbitrary adaptive coefficients can contain the fresh primes. The next local inequality quantifies the price.

Proposition 4.5 (Cyclotomic coefficient-or-denominator tax). *Assume that ℓ is larger than every prime divisor of MA . Let $S = \sum_{k=1}^\ell a_k J_k(r, s)$ with $a_\ell \neq 0$. For $p^d \parallel \mathfrak{F}_\ell$ put*

$$C_p = \max_{\substack{1 \leq k \leq \ell \\ a_k \neq 0}} |v_p(a_k)|, \quad E_p = \max\{0, -v_p(S)\}.$$

Then

$$E_p + C_p \geq \frac{d}{2}. \quad (19)$$

Consequently,

$$\log \text{den}(S) + \sum_{p \mid \mathfrak{F}_\ell} C_p \log p \geq \frac{1}{2} \log \mathfrak{F}_\ell, \quad (20)$$

and, writing $h(a)$ for rational logarithmic height with $h(0) = 0$,

$$\boxed{\log \text{den}(S) + 2 \sum_{k=1}^\ell h(a_k) \geq \frac{1}{2} \log \mathfrak{F}_\ell.} \quad (21)$$

Proof. If $C_p \geq d/2$, there is nothing to prove. Suppose $C_p < d/2$. By [Theorem 4.3](#), the top summand has valuation $v_p(a_\ell) - d < -d/2$, whereas every lower summand has valuation at least $-C_p > -d/2$. The top summand is therefore the unique term of smallest valuation, so

$$E_p = d - v_p(a_\ell) \geq d - C_p > d/2.$$

This proves (19). Sum over p . Finally,

$$\sum_p |v_p(a)| \log p = \log |\text{num } a| + \log \text{den } a \leq 2h(a),$$

and $C_p \leq \sum_k |v_p(a_k)|$. $\square$

The factor $1/2$ is the correct local scale for this elementary dichotomy: a coefficient can partly neutralize the top denominator while another coefficient introduces a matching negative valuation. The conclusion is not that large coefficients are impossible, but that a proposed cancellation mechanism must visibly carry the same moving prime mass that it removes.

5 Exact maximal-contact stencils

Put

$$f_x(z) = \frac{\sinh(xz)}{\sinh z}, \quad f_x(0) = x. \quad (22)$$

Let

$$1 \leq n_1 < \cdots < n_m = N$$

be distinct positive integer nodes and set $\lambda_i = n_i^2$. The unique weights that reproduce the value at zero for every polynomial in z^2 of degree at most $m-1$ are

$$w_i = \prod_{j \neq i} \frac{\lambda_j}{\lambda_j - \lambda_i} = \prod_{j \neq i} \frac{n_j^2}{(n_j - n_i)(n_j + n_i)}. \quad (23)$$

Define the rational stencil

$$\mathcal{S}_{\mathbf{n},x}(e^h) = \sum_{i=1}^m w_i f_x(n_i h) = \sum_{i=1}^m w_i J_{n_i}(r, s), \quad r = e^h. \quad (24)$$

Under (2) and for a 2–3 rational r , every term is rational.

Theorem 5.1 (Exact moment-generating identity). *For every $j \geq 0$ let $M_j = \sum_{i=1}^m w_i \lambda_i^j$. Then*

$$M_0 = 1, \quad M_j = 0 \quad (1 \leq j < m), \quad M_m = (-1)^{m-1} \prod_{i=1}^m \lambda_i, \quad (25)$$

and, as a rational-function identity,

$$\boxed{\sum_{j \geq 0} (M_j - \mathbf{1}_{j=0}) t^j = \frac{(-1)^{m-1} (\prod_i \lambda_i) t^m}{\prod_i (1 - \lambda_i t)}}. \quad (26)$$

If $f_x(z) = x + \sum_{j \geq 1} b_j(x) z^{2j}$, then

$$\mathcal{S}_{\mathbf{n},x}(e^h) - x = (-1)^{m-1} \left(\prod_i \lambda_i \right) h^{2m} \sum_{r \geq 0} b_{m+r}(x) h^{2r} h_r(\lambda_1, \dots, \lambda_m), \quad (27)$$

where h_r is the complete homogeneous symmetric polynomial. The identity (27) is formal in h ; as an analytic series it is valid whenever $N|h| < \pi$.

Proof. The first m identities are the Lagrange interpolation identities at zero for $1, t, \dots, t^m$. Alternatively,

$$\sum_{i=1}^m \frac{w_i}{1 - \lambda_i t} - 1$$

has denominator $\prod_i (1 - \lambda_i t)$ and a zero of order m at $t = 0$. Its numerator has degree at most m , and comparison of the t^m coefficient gives (26). Expanding the reciprocal product gives (27). $\square$

Lemma 5.2 (Prime support of maximal-contact weights). *Every prime occurring in the numerator or denominator of a weight w_i in (23) is strictly smaller than $2N$.*

Proof. The displayed factorization of w_i uses only the positive integers $n_j < N+1$, $|n_j - n_i| < N$, and $n_j + n_i < 2N$. Reduction of the rational number cannot create a new prime. $\square$

Corollary 5.3 (Prime-top survival for every maximal-contact stencil). *Assume that the largest node N is an odd prime and is larger than every prime divisor of MA . Then*

$$\mathfrak{F}_N(P, Q; U, V) \mid \text{den } \mathcal{S}_{\mathbf{n},x}(e^h), \quad (28)$$

and hence

$$\log \text{den } \mathcal{S}_{\mathbf{n},x}(e^h) \geq (N-1) \log P - o_x(N \log P). \quad (29)$$

Proof. Every prime in $\mathfrak{F}_N$ is larger than $2N$ by [Theorem 4.1](#), while every weight is a unit at such a prime by [Theorem 5.2](#). Apply [Theorem 4.4](#) with zero coefficients at the unused levels. $\square$

This corollary closes an entire class at once: changing the integer nodes does not avoid the top cyclotomic factor as long as the stencil is maximally flat and its largest node is prime.

5.1 The consecutive-node Richardson filter as a central difference

For $n_i = i$ ($1 \leq i \leq m$), the weights become

$$c_{m,k} = (-1)^{k-1} \frac{2(m!)^2}{(m-k)!(m+k)!} = \frac{2(-1)^{k-1}}{\binom{2m}{m}} \binom{2m}{m+k}. \quad (30)$$

Write

$$\mathcal{R}_{m,x}(e^h) = \sum_{k=1}^m c_{m,k} f_x(kh).$$

For an even function f , define the centered difference

$$\delta_h^{2m} f(0) = \sum_{k=-m}^m (-1)^{m+k} \binom{2m}{m+k} f(kh). \quad (31)$$

Proposition 5.4 (Exact central-difference identity). *One has*

$$\boxed{\mathcal{R}_{m,x}(e^h) - x = \frac{(-1)^{m+1}}{\binom{2m}{m}} \delta_h^{2m} f_x(0).} \quad (32)$$

Equivalently, for every complex t ,

$$\boxed{\sum_{k=1}^m c_{m,k} \cosh(kt) = 1 - \frac{(-1)^m}{\binom{2m}{m}} (2 \sinh(t/2))^{2m}.} \quad (33)$$

Proof. Pair the k and $-k$ terms in (31), use evenness, and compare with (30). For (33), apply the resulting identity to $f(z) = \cosh(tz/h)$ and use

$$\sum_{k=-m}^m (-1)^{m+k} \binom{2m}{m+k} e^{kt} = (e^{t/2} - e^{-t/2})^{2m}.$$

$\square$

Proposition 5.5 (Exact calibration on integer exponents). *For a positive integer L ,*

$$\mathcal{R}_{m,L}(e^h) - L = -\frac{2(-1)^m}{\binom{2m}{m}} \sum_{j=1}^{(L-1)/2} (2 \sinh(jh))^{2m}, \quad L \text{ odd}, \quad (34)$$

$$\mathcal{R}_{m,L}(e^h) - L = -\frac{2(-1)^m}{\binom{2m}{m}} \sum_{j=1}^{L/2} (2 \sinh((2j-1)h/2))^{2m}, \quad L \text{ even}. \quad (35)$$

Proof. Use the finite Fourier expansions

$$\frac{\sinh(Lz)}{\sinh z} = 1 + 2 \sum_{j=1}^{(L-1)/2} \cosh(2jz)$$

for odd L , and

$$\frac{\sinh(Lz)}{\sinh z} = 2 \sum_{j=1}^{L/2} \cosh((2j-1)z)$$

for even L , then apply (33) term by term. $\square$

The exact formulas distinguish the integral locus from the pole-dominated nonintegral regime below. They are also compact targets for formalization.

6 Variable-order asymptotics from the nearest poles

Let

$$f_x(z) = x + \sum_{j \geq 1} b_j(x) z^{2j}. \quad (36)$$

For nonintegral x , the nearest poles of f_x are at $z = \pm i\pi$.

Lemma 6.1 (Nearest-pole coefficient asymptotic). *Fix $x \notin \mathbb{Z}$. For every ρ with $\pi < \rho < 2\pi$,*

$$b_j(x) = \frac{2(-1)^j \sin(\pi x)}{\pi^{2j+1}} + O_{x,\rho}(\rho^{-2j}). \quad (37)$$

Proof. The sum of the principal parts at $\pm i\pi$ is

$$\frac{2\pi \sin(\pi x)}{z^2 + \pi^2}.$$

Subtract it from f_x . The difference is holomorphic in every closed disk $|z| \leq \rho < 2\pi$. Cauchy's coefficient estimate gives the error term, while expansion of the displayed rational function gives the main term. $\square$

Theorem 6.2 (Variable-node pole product). *Fix $x \notin \mathbb{Z}$. Let $m \rightarrow \infty$, let $1 \leq n_1 < \dots < n_m = N$, and let $h \rightarrow 0$ through real values such that*

$$N|h| \rightarrow 0. \quad (38)$$

Then the maximal-contact stencil (24) satisfies

$$\boxed{\mathcal{S}_{\mathbf{n},x}(e^h) - x = -\frac{2\sin(\pi x)}{\pi} \prod_{i=1}^m \frac{n_i^2 h^2}{\pi^2 + n_i^2 h^2} (1 + o_x(1))}. \quad (39)$$

In particular, the error is eventually nonzero and has the fixed sign $-\operatorname{sgn}(\sin \pi x)$.

Proof. Insert (37) into the exact tail (27). The contribution of the two nearest poles is

$$\begin{aligned} & -\frac{2\sin(\pi x)}{\pi} \left(\prod_i \frac{n_i^2 h^2}{\pi^2} \right) \sum_{r \geq 0} (-1)^r h_r(\lambda_1, \dots, \lambda_m) \left(\frac{h^2}{\pi^2} \right)^r \\ & = -\frac{2\sin(\pi x)}{\pi} \prod_i \frac{n_i^2 h^2}{\pi^2 + n_i^2 h^2}. \end{aligned}$$

For a fixed $\rho \in (\pi, 2\pi)$, the contribution of the remainder is at most a constant times

$$\left(\prod_i \frac{n_i^2 h^2}{\rho^2} \right) \prod_i \left(1 - \frac{n_i^2 h^2}{\rho^2} \right)^{-1}.$$

After division by the main term, its logarithm is bounded above by

$$-2m \log(\rho/\pi) + O\left(h^2 \sum_i n_i^2\right) = -2m \log(\rho/\pi) + o(m),$$

because $h^2 \sum_i n_i^2 \leq mN^2 h^2 = o(m)$. The relative error therefore tends to zero exponentially in m . $\square$

For consecutive nodes, [Theorem 6.2](#) gives the more explicit formula

$$\mathcal{R}_{m,x}(e^h) - x = -\frac{2 \sin(\pi x)}{\pi} (m!)^2 \left(\frac{h}{\pi}\right)^{2m} \prod_{k=1}^m \left(1 + \frac{k^2 h^2}{\pi^2}\right)^{-1} (1 + o_x(1)). \quad (40)$$

If the stronger condition $m^3 h^2 \rightarrow 0$ holds, the product is $1 + o(1)$ and

$$\mathcal{R}_{m,x}(e^h) - x = -\frac{2 \sin(\pi x)}{\pi} (m!)^2 \left(\frac{h}{\pi}\right)^{2m} (1 + o_x(1)). \quad (41)$$

Stirling's formula then yields

$$-\log |\mathcal{R}_{m,x}(e^h) - x| = 2m \log \frac{\pi e}{m|h|} + O_x(\log m). \quad (42)$$

7 The contact-height mismatch is now quantitative

Let p_j/q_j be continued-fraction convergents to $\log 3/\log 2$, and put

$$h_j = q_j \log 3 - p_j \log 2. \quad (43)$$

Replace (p_j, q_j) by the opposite direction when necessary so that $r_j = e^{|h_j|} > 1$, and write $r_j = P_j/Q_j$ in lowest terms. Then

$$\log P_j \asymp q_j, \quad |h_j| \ll q_j^{-1}. \quad (44)$$

A lower bound for a nonzero linear form in $\log 2$ and $\log 3$ gives constants $C_0, C_1 > 0$ such that

$$|h_j| \geq C_0 q_j^{-C_1}, \quad \log(1/|h_j|) = O(\log q_j) = o(\log P_j). \quad (45)$$

This weak-looking logarithmic comparison is decisive once the denominator has cyclotomic scale $N \log P_j$.

Theorem 7.1 (Richardson route closed in the growing prime window). *Let ℓ_j be odd primes with*

$$\ell_j \rightarrow \infty, \quad \ell_j |h_j| \rightarrow 0. \quad (46)$$

For the rational Richardson approximants $R_j = \mathcal{R}_{\ell_j, x}(e^{|h_j|})$, one has

$$\boxed{\frac{-\log |R_j - x|}{\log \text{den}(R_j)} \rightarrow 0.} \quad (47)$$

Equivalently, for every fixed $\eta > 0$,

$$|R_j - x| > \text{den}(R_j)^{-\eta} \quad (48)$$

for all sufficiently large j .

Proof. The weights (30) have prime support at most $2\ell_j$, so Theorems 4.2 and 4.4 give

$$\log \text{den}(R_j) \geq (1 - o(1))\ell_j \log P_j.$$

The pole product (40) and $\ell_j|h_j| \rightarrow 0$ give

$$-\log |R_j - x| \leq 2\ell_j \log \frac{C}{|h_j|} + o(\ell_j)$$

after discarding terms that only reduce the negative logarithm. Divide and use (45). $\square$

A concrete admissible choice is any prime $\ell_j \asymp q_j^{1/2}$, whose existence follows from Bertrand's postulate. The theorem is stronger than the fixed-order observation: the order tends to infinity and may grow almost as fast as $1/|h_j|$.

Theorem 7.2 (All prime-top maximal-contact integer stencils are weak). *For each j , let*

$$1 \leq n_{j,1} < \cdots < n_{j,m_j} = N_j$$

be integer nodes with $m_j \rightarrow \infty$, where N_j is an odd prime. Assume

$$N_j|h_j| \rightarrow 0. \tag{49}$$

Let S_j be the corresponding maximal-contact rational stencil (24). Then

$$\boxed{\frac{-\log |S_j - x|}{\log \text{den}(S_j)} \rightarrow 0.} \tag{50}$$

Proof. By Theorem 5.3,

$$\log \text{den}(S_j) \geq (1 - o(1))N_j \log P_j.$$

By Theorem 6.2,

$$\begin{aligned} -\log |S_j - x| &= 2 \sum_{i=1}^{m_j} \log \frac{\sqrt{\pi^2 + n_{j,i}^2 h_j^2}}{n_{j,i} |h_j|} + O_x(1) + o(1) \\ &\leq 2m_j \log \frac{C}{|h_j|} + o(m_j) \leq 2N_j \log \frac{C}{|h_j|} + o(N_j). \end{aligned}$$

Divide and apply (45). $\square$

Interpretation. A maximally flat m -node even stencil cannot evade the height problem by moving its nodes. Prime-top cyclotomic isolation contributes denominator mass of order $N \log P$, while the real contact supplies at most order $m \log(1/|h|)$. Since $m \leq N$ and Baker gives $\log(1/|h|) = o(\log P)$, the approximation exponent tends to zero. This closes the natural class of linear, integer-node, maximal-contact extrapolants in the near-identity regime $N|h| \rightarrow 0$.

The theorem does not cover nonlinear constructions, stencils whose largest index lies outside the near-identity window, or coefficient systems not chosen by maximal polynomial contact. Those are genuine remaining options, not loopholes inside the proved statement.

8 An adelic covolume conservation law

A different proposed route is to force approximation denominators to carry large powers of selected primes and then combine real and p -adic smallness. The basic geometry-of-numbers architecture has an exact budget identity.

Fix a finite set of primes S and exponents $e_p \geq 0$. Put

$$L = \prod_{p \in S} p^{e_p}. \quad (51)$$

For a real number β , consider the lattice

$$\Lambda_{\beta,L} = \{(q, q\beta - a) : q \in L\mathbb{Z}, a \in \mathbb{Z}\} \subset \mathbb{R}^2. \quad (52)$$

Theorem 8.1 (Covolume conservation). *The lattice (52) has covolume L . If $Q, E > 0$, $E < 1$, and*

$$QE > L, \quad (53)$$

then there are integers $q \neq 0, a$ such that

$$q \in L\mathbb{Z}, \quad |q| \leq Q, \quad |q\beta - a| \leq E. \quad (54)$$

For every such q ,

$$\prod_{p \in S} |q|_p \leq L^{-1}. \quad (55)$$

Thus the guaranteed scale furnished by Minkowski is

$$|q\beta - a| \prod_{p \in S} |q|_p \lesssim Q^{-1}, \quad (56)$$

exactly the unrestricted Dirichlet exponent.

Proof. A basis for $\Lambda_{\beta,L}$ is $(L, L\beta)$ and $(0, -1)$, whose determinant has absolute value L . The symmetric box $[-Q, Q] \times [-E, E]$ has area $4QE > 4L$. Minkowski’s convex body theorem therefore supplies a nonzero lattice point. Because $E < 1$, a point with $q = 0$ would have $a = 0$, so the point has $q \neq 0$. This proves (54). Divisibility by L gives $v_p(q) \geq e_p$ and hence (55). Choosing $E \asymp L/Q$ cancels the forced local factor L^{-1} . $\square$

Proposition 8.2 (Sharpness of the real scale). *For each fixed L , there are real β and a constant $c > 0$ such that all $q \in L\mathbb{Z} \setminus \{0\}$ and $a \in \mathbb{Z}$ satisfy*

$$|q\beta - a| \geq \frac{cL}{|q|}. \quad (57)$$

Hence no uniform improvement on the scale L/Q is possible in this restricted-denominator architecture.

Proof. Choose a badly approximable real number α and set $\beta = \alpha/L$. For $q = Lm$, $|q\beta - a| = |m\alpha - a| \geq c/|m| = cL/|q|$. $\square$

The covolume law is not a universal impossibility theorem for adelic approximation. It rules out only the direct strategy “impose $q \equiv 0 \pmod{L}$ and invoke Minkowski.” Extra accidental p -adic divisibility, correlations with the numerator $q\beta - a$, higher-dimensional determinant constructions, or output-specific congruences are outside its scope.

9 A tensor-period interface and the Drinfeld comparison

The logarithmic determinant attached to (2) is

$$\Delta_x = \log M \log 3 - \log A \log 2 = 0. \quad (58)$$

Baker’s theorem controls algebraic linear combinations of logarithms of algebraic numbers, but (58) is quadratic in those logarithms. Logarithms of algebraic numbers are periods of Kummer 1-motives. The period conjecture for 1-motives is known through the analytic subgroup theorem, and recent work gives explicit dimension formulas for their spaces of linear periods [7, 6]. However, products such as $\log M \log 3$ live naturally in a tensor product; the category of 1-motives is not closed under the tensor operation needed to keep (58) inside the known linear theory.

This identifies a precise target rather than a vague appeal to motives.

Research target 9.1 (Tensor-period lift). Construct a controlled tensor-period object generated by the Kummer data of $2, 3, M, A$ for which (58) becomes a linear period relation, and prove that every such relation is induced by explicit multiplicative morphisms. Show that the relation (58) then forces $x \in \mathbb{Q}$.

In the function-field setting, Frobenius difference equations and t-motivic Galois groups do provide algebraic-independence theorems for periods and logarithms of Drinfeld modules. For example, Chang and Papanikolas prove algebraic independence results for logarithms and quasi-logarithms of rank-two Drinfeld modules without complex multiplication [3]. The comparison suggests that the missing classical ingredient is not another estimate for a single linear form, but an operator or Tannakian symmetry that turns rank drop in a logarithm matrix into a controlled linear relation in a tensor category. At present this is a research direction, not a theorem applicable to (1).

10 Exact computational stress tests

The cyclotomic survival theorem is asymptotic, but its local mechanism can be checked exactly for arbitrary multiplicatively independent rational pairs. These pairs are *not* hypothetical Alaoglu–Erdős candidates; they are unit tests for the valuation argument. For each pair we computed the consecutive-node Richardson stencil exactly with Python’s `Fractions.Fraction` and verified

$$\mathfrak{F}_\ell \mid \text{den } \mathcal{R}_\ell.$$

r	s	ℓ	$\Phi_{2\ell}$	gcd	$\mathfrak{F}_\ell$	check
3/2	5/4	7	463	1	463	divides
3/2	5/4	11	35839	1	35839	divides
3/2	5/4	13	320503	79	4057	divides
3/2	5/4	17	25854247	1	25854247	divides
9/8	10/9	5	5401	11	491	divides
9/8	10/9	7	404713	1	404713	divides
9/8	10/9	11	2351234953	1	2351234953	divides
4/3	7/5	5	181	1	181	divides
4/3	7/5	13	9814741	313	31357	divides
4/3	7/5	19	39434309773	1	39434309773	divides

The cases with nontrivial gcd show why one must remove target-gap cancellation before claiming survival. The remaining factor is still found in the reduced denominator of the complete sum, exactly as [Theorem 4.4](#) predicts.

10.1 Reproducible exact-arithmetic script

```

1 from fractions import Fraction
2 from math import factorial, gcd
3
4
5 def coeff(m, k):
6     return Fraction((((-1) ** (k - 1)) * 2 * factorial(m) ** 2,
7                     factorial(m - k) * factorial(m + k))
8
9
10 def J(P, Q, U, V, k):
11     return Fraction((U ** (2*k) - V ** (2*k)) * (P*Q) ** k,
12                    (P ** (2*k) - Q ** (2*k)) * (U*V) ** k)
13
14
15 def richardson(P, Q, U, V, m):
16     return sum((coeff(m, k) * J(P, Q, U, V, k)
17                for k in range(1, m + 1)), Fraction(0, 1))
18
19
20 def phi_2prime(P, Q, ell):
21     return (P ** ell + Q ** ell) // (P + Q)
22
23
24 def remove_ell_part(n, ell):
25     while n % ell == 0:
26         n //= ell
27     return n
28
29
30 def fresh_factor(P, Q, U, V, ell):
31     phi = phi_2prime(P, Q, ell)
32     cancelled = gcd(phi, U ** (2*ell) - V ** (2*ell))
33     return remove_ell_part(phi // cancelled, ell)
34
35
36 for P, Q, U, V in [(3, 2, 5, 4), (9, 8, 10, 9), (4, 3, 7, 5)]:
37     for ell in [3, 5, 7, 11, 13, 17, 19]:
38         F = fresh_factor(P, Q, U, V, ell)
39         R = richardson(P, Q, U, V, ell)
40         assert R.denominator % F == 0

```

11 Lean-oriented formalization plan

The new argument naturally separates finite algebra from external analytic and Diophantine inputs.

11.1 Kernel-friendly finite modules

1. **Lagrange moments.** Formalize [\(23\)](#), $M_0 = 1$, $M_j = 0$ for $1 \leq j < m$, and the rational identity [\(26\)](#). This is finite-field/rational-function algebra.
2. **Central difference.** Prove [\(32\)](#) and [\(33\)](#) from the binomial theorem.

3. **Integer calibration.** Prove the finite hyperbolic expansions and (34)–(35).
4. **Cyclotomic order.** For odd prime ℓ , prove that a prime $p \neq \ell$ dividing $\Phi_{2\ell}(P, Q)$ has order 2ℓ , and prove the LTE bound $v_\ell \leq 1$.
5. **Unique negative valuation.** Formalize [Theorem 4.3](#) and the ultrametric one-term survival lemma.
6. **Coefficient tax.** Formalize the local dichotomy (19); this is an ordered-group argument on $\mathbb{Z} \cup \{\infty\}$.
7. **Weight support.** Prove that primes in the Lagrange weights are less than $2N$, giving [Theorem 5.3](#) once the cyclotomic input is available.
8. **Covolume.** Formalize the basis determinant of $\Lambda_{\beta, L}$ and invoke an existing Minkowski theorem when available.

11.2 External theorem interfaces

Three inputs should be represented as explicit assumptions or imported library theorems rather than silently re-proved:

1. the fixed-group Corvaja–Zannier generalized-gcd theorem;
2. a lower bound for $|q \log 3 - p \log 2|$ of polynomial type in q ;
3. Cauchy coefficient bounds for a meromorphic function after subtracting its two principal parts.

A useful dependency order is

$$\begin{aligned} \text{LagrangeMoments} &\longrightarrow \text{PrimeTopSupport} \longrightarrow \text{CyclotomicIsolation} \\ &\longrightarrow \text{ReducedStencilDenominator}, \end{aligned}$$

with the analytic branch

$$\begin{aligned} \text{NearestPoleCoefficients} &\longrightarrow \text{PoleProductAsymptotic} \\ &\longrightarrow \text{ContactHeightNoGo}. \end{aligned}$$

The final no-go theorem then imports the two external estimates and joins the two branches. None of these modules should be labeled as a formal proof of the original conjecture.

12 What remains viable

The new results substantially narrow, but do not eliminate, the search.

12.1 Nonlinear constructions

[Theorem 7.2](#) covers linear combinations of rational symmetric secants with maximal polynomial contact. A viable construction must use nonlinear combinations, determinants with essential cross-level cancellation, or a functional equation that changes the prime-order support rather than merely reweighting it. Any proposal should be tested first against the unique-top-valuation mechanism of [Theorem 4.3](#).

12.2 Output-specific cancellation beyond the fixed-group theorem

The orbit-uniform gcd theorem proves that direct cancellation between $\Phi_{2\ell}(P, Q)$ and $U^{2\ell} - V^{2\ell}$ has only $o(\ell \log P)$ mass. A successful finite-place proof therefore needs a more structured overlap involving several target gaps or conjugates, not a single pair. The missing statement would resemble a quantitative local-to-global theorem for a family of correlated cyclotomic factors.

12.3 Tensor periods

A theorem controlling linear relations among products of logarithmic periods could attack (58) directly. The precise obstacle is that known 1-motive period results are linear, while the determinant is a relation in a tensor square. The Drinfeld-module analogy shows what an operator-driven solution would look like, but no classical theorem currently bridges this gap in the required generality.

12.4 Outside the near-identity window

The pole-product theorem assumes $N|h| \rightarrow 0$. If $N|h|$ is bounded away from zero, the sample points $n_i h$ no longer lie in one shrinking analytic neighborhood, and maximal Taylor contact is not the correct language. This regime could support a genuinely different spectral or saddle-point construction. It also has larger numerator and denominator growth, so a proposal must include a complete height calculation rather than only an analytic smallness estimate.

Recommended next target. The highest-value concrete problem is to prove or disprove the following: for every rational nonlinear expression of bounded algebraic degree in the prime-top values $J_1, \dots, J_\ell$, either a positive proportion of $\log \Phi_{2\ell}(P, Q)$ survives in its reduced height, or the expression factors through an explicit multiplicative dependence. This would extend cyclotomic isolation from linear stencils to a controlled nonlinear class and could either produce the missing contradiction or reveal the exact algebraic shape a counterexample would require.

13 Conclusion

The work reported here does not settle the Alaoglu–Erdős conjecture, but it produces several nontrivial upgrades that are independent of the supplied report’s main constructions.

The fixed-group observation turns Corvaja–Zannier into a moving-direction tool. That uniformity makes it possible to isolate almost the entire prime-order cyclotomic factor. The factor then survives reduction of the whole maximal-contact stencil, not merely one unreduced summand. Exact moment-generating and central-difference identities expose the analytic side of the same construction, and the nearest poles give a variable-order product formula. Baker’s logarithmic separation shows that the real contact budget is only logarithmic in the direction height, whereas the surviving cyclotomic denominator budget is linear in that height. Consequently, a large and natural class of growing linear extrapolants is rigorously too weak. Finally, the adelic covolume calculation shows why simply imposing local denominator depth cannot restore the missing exponent.

The proof search should now concentrate on mechanisms that are genuinely nonlinear, tensorial, or multi-gap. Repeating fixed-order interpolation, adding more integer nodes, or forcing prescribed denominator divisibility without a new correlation cannot close the conjecture.

A A local valuation template for future filters

The proof of [Theorem 4.5](#) is useful beyond the present stencil. Let K be a nonarchimedean field with valuation v , and suppose $b_0, \dots, b_m$ satisfy

$$v(b_m) = -d < 0, \quad v(b_i) \geq 0 \quad (i < m).$$

For nonzero coefficients a_i and $S = \sum a_i b_i$, put $C = \max_i |v(a_i)|$ and $E = \max(0, -v(S))$. Then $E + C \geq d/2$. This statement is completely independent of cyclotomic polynomials. Any future construction that identifies a prime at which one level has a unique negative valuation immediately inherits the same coefficient-or-denominator tax.

B Bibliographic notes

The use of Corvaja–Zannier in [Theorem 3.1](#) is qualitative but uniform in the fixed finitely generated group; no effective exceptional height is claimed. For (45), any standard effective lower bound for a nonzero linear form in two logarithms suffices; the precise constant is irrelevant. The period discussion is deliberately diagnostic: it records the categorical level at which the known theorem stops and does not assert a new motivic independence result.

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